

PUMaC 2017 Power Round:

“I ain’t even Lie-in’...”

November 17, 2017

“Out of nothing I have created a strange new universe.” —János Bolyai

Rules and Reminders

1. Your solutions may be turned in in one of two ways:
 - You may email them to us at **pumac2017power@gmail.com** by 8AM Eastern Standard Time on the morning of PUMaC, November 18, 2017 with the subject line “PUMaC 2017 Power Round.”
 - You may hand them in to us when your team checks in on the morning of PUMaC. Please staple your solutions together, including the cover sheet.

The cover sheet (the last page of this document) should be the first page of your submission. Each page should have on it the **team number** (not team name) and **problem number**. This number can be found by logging in to the coach portal and selecting the corresponding team. Solutions to problems may span multiple pages, but include them in continuing order of proof.

2. You are encouraged, but not required, to use $\text{\LaTeX}$ to write your solutions. If you submit your power round electronically, **you may not submit multiple times**. The first version of the power round solutions that we receive from your team will be graded. If submitting electronically, **you must submit a PDF**. No other file type will be graded.
3. Do not include identifying information aside from your team number in your solutions.
4. Please collate the solutions in order in your solution packet. Each problem should start on a new page, and solutions should be written on one side of the paper only (there is a point deduction for not following this formatting).
5. On any problem, you may use without proof any result that is stated earlier in the test, as well as any problem from earlier in the test, even if it is a problem that your team has not solved. You may not cite parts of your proof of other problems: if you wish to use a lemma in multiple problems, please reproduce it in each one.

6. When a problem asks you to “show,” “prove” or “demonstrate” a result, a formal proof is expected, in which you justify each step you take, either by using a method from earlier or by *proving* that *everything* you do is correct. When a problem instead uses the word “explain,” an informal explanation suffices. When a problem asks you to “find” or “list” something, no justification is required.
7. All problems are numbered as “Problem x.y.z” where x is the section number and y is the subsection. Each problem’s point distribution can be found in parentheses before the problem statement.
8. **You may NOT use any references, such as books or electronic resources, unless otherwise specified. You may NOT use computer programs, calculators, or any other computational aids.**
9. Teams whose members use English as a foreign language may use dictionaries for reference.
10. **Communication with humans outside your team of 8 students about the content of these problems is prohibited.**
11. There are two places where you may ask questions about the test. The first is Piazza. Please ask your coach for instructions to access our Piazza forum. On Piazza, you may ask any question **so long as it does not give away any part of your solution to any problem**. If you ask a question on Piazza, all other teams will be able to see it. If such a question reveals all or part of your solution to a power round question, your team’s power round score will be penalized severely. For any questions you have that might reveal part of your solution, or if you are not sure if your question is appropriate for Piazza, please email us at pumac@math.princeton.edu. We will email coaches with important clarifications that are posted on Piazza.

Introduction and Advice

The topic of this power round is **Lie algebras** (“Lie” pronounced as “Lee”). Lie algebras are mathematical objects with a simple set of operations that leads to a powerful classification. Rather than being a trivial matter untouched by modern mathematicians, Lie algebras are integral to a great deal of cutting-edge research; just last year, mathematicians fully cracked the structure of the algebra corresponding to the E_8 Dynkin diagram (shown on page 20).

Sections 1 and 2 introduce fundamental ideas of linear algebra. Sections 3 and 4 provide some theory of Lie algebras through a problem-solving approach. Section 5 it discusses graphs known as Dynkin diagrams and works through elegant cases of Serre’s Theorem.

This is not intended to be a complete course in Lie algebras; in any event, a contest is far from the best way to provide a complete undertaking. Rather, think of this as a groundwork for Section 5, which contains some truly beautiful results, in a sort of “greatest hits” of linear algebra.

Instead of having very few problems with many steps apiece, the guiding philosophy behind Sections 3-5 is that the majority of the problems are intended to be solved using only a handful of leaps. This is meant to reward understanding, as the progression in these sections is meant to be incremental.

Here is some further advice with regard to the Power Round:

- **Read the text of every problem!** Many important ideas are included in problems and may be referenced later on. In addition, some of the theorems you are asked to prove are useful or even necessary for later problems.
- **Make sure you understand the definitions**, especially in the last few sections. If you don’t, then you will not be able to do the problems. Feel free to ask clarifying questions about the definitions on Piazza (or email us).
- **Don’t make stuff up:** on problems that ask for proofs, but you will receive more points if you demonstrate legitimate and correct intuition than if you fabricate something that *looks* rigorous just for the sake of having “rigor.”
- **Check Piazza often!** Clarifications will be posted there, and if you have a question it is possible that it has already been asked and answered in a Piazza thread (and if not, you can ask it, assuming it does not reveal any part of your solution to a question). **If in doubt about whether a question is appropriate for Piazza, please email us at pumac@math.princeton.edu.**

Good luck, and have fun!

– Zachary Stier

We’d like to acknowledge and thank many individuals and organizations for their support; without their help, this Power Round (and the entire competition) could not exist. Please refer to the solution of the power round for full acknowledgments.

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Notation

- $\forall$: for all. *ex.*: $\forall x \in \{1, 2, 3\}$ means “for all x in the set $\{1, 2, 3\}$ ”
- $f \circ g$: function composition. *ex.*: $(f \circ g)(x) = f(g(x))$.
- $A \subset B$: proper subset. *ex.*: $\{1, 2\} \subset \{1, 2, 3\}$, but $\{1, 2\} \not\subset \{1, 2\}$
- $A \subseteq B$: subset, possibly improper. *ex.*: $\{1\}, \{1, 2\} \subseteq \{1, 2\}$
- $f : x \mapsto y$: f maps x to y . *ex.*: if $f(n) = n - 3$ then $f : 20 \mapsto 17$ and $f : n \mapsto n - 3$ are both true.
- $\{x \in S \mid C(x)\}$: the set of all x in the set S satisfying the condition $C(x)$. *ex.*: $\{n \in \mathbb{N} \mid \sqrt{n} \in \mathbb{N}\}$ is the set of perfect squares.
- $\mathbb{N}$: the natural numbers $\{1, 2, 3, \dots\}$.
- $\mathbb{R}$: the real numbers.
- $\mathbb{C}$: the complex numbers, $\{x + iy \mid x, y \in \mathbb{R} \text{ and } i = \sqrt{-1}\}$.

0 Whitelist

Linear algebra can be tough to grasp in its first go. For teams that might like a more in-depth look at certain aspects, **only** the following resources may be referenced to enhance understanding:

- Robert Beezer, *A First Course in Linear Algebra*, <http://linear.ups.edu/fcla/index.html> – I would recommend this link first, as it appears to be an intermediate stage between the exposition of this Power Round and the following two texts.
- Jim Hefferon, *Linear Algebra*, 3 ed., <http://joshua.smcvt.edu/linearalgebra/book.pdf>
- David Cherney et al., *Linear Algebra*, <https://www.math.ucdavis.edu/~linear/linear-guest.pdf>

These are the only resources you may access while working on the Power Round. Using any other resource constitutes cheating.

In addition, you may only **cite the texts when those results are consistent with only what has been featured prior to that point in the Power Round**. This is to ensure that the texts are purely supplemental to the Round.

What follows is a brief crash course in linear algebra, which will begin to lay down the foundations for the crux of this Power Round. If it appears that the coming sections are rather disjointed, understand that a great deal of content has been omitted as to only include the topics essential for our course of study.

1 Linear Algebra I (20 points)

1.1 Vector Spaces (5 points)

Definition 1.1.A. A *vector space* is a set V of elements (known as *vectors*) that is associated with a *field* F . In what follows, F will be either $\mathbb{R}$ or $\mathbb{C}$ (though these are not the only fields).

- A vector space is closed under two operations:
 - addition ($\mathbf{v} + \mathbf{w}$ is an element of V for any $\mathbf{v}, \mathbf{w} \in V$)
 - *scalar multiplication* ($a\mathbf{v}$ is an element of V for any $a \in F$ and $\mathbf{v} \in V$)
- There is a *zero element* $0 \in V$ that functions as an additive identity.
- Scalar multiplication distributes over vector addition ($a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$) and vice-versa ($(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$), plus we have the following associativity: $a(b\mathbf{v}) = (ab)\mathbf{v}$.

Note that this implies the existence of additive inverses in V , i.e. for any $\mathbf{v} \in V$ there is another element, $\mathbf{w}$, for which $\mathbf{v} + \mathbf{w} = 0$.

Two illustrative initial examples are $\mathbb{R}$ and $\mathbb{C}$, which are each vector spaces over themselves. We see that each is closed under addition and scalar multiplication (which is actually just multiplication distributing over addition), and multiplying by -1 gives additive inverses. $\mathbb{R}^n$ and $\mathbb{C}^n$, the ordered n -tuples $(a_1, a_2, \dots, a_n)$ of real or complex numbers, are for the same reasons vector spaces.

Definition 1.1.B. $W \subseteq V$ is a *subspace* if W is itself a vector space.

For instance, $\{(z, 0) \mid z \in \mathbb{C}\} \subset \mathbb{C}^2$ is a subspace relation.

If two vector spaces are “independent from each other,” in that their only common element is 0, then we write $V_1 \oplus V_2 = \{\mathbf{v}_1 + \mathbf{v}_2 \mid \mathbf{v}_1 \in V_1, \mathbf{v}_2 \in V_2\}$. For instance, if $V_1 = \{(r, 0) \mid r \in \mathbb{R}\}$ and $V_2 = \{(0, r) \mid r \in \mathbb{R}\}$ then $V_1 \oplus V_2 = \mathbb{R}^2$. This lends itself to a notion of “adding” disjoint vector spaces.

Definition 1.1.C. A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is *linearly independent* if there are no $c_1, c_2, \dots, c_n \in F$ such that not all of them are 0 and $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = 0$.

For instance, $(2, 0), (1, 7) \in \mathbb{R}^2$ are linearly independent because they cannot be combined to 0. However, no two nonzero real numbers are linearly independent.

Definition 1.1.D. A subset $V' \subset V$ *spans* V if, given $\mathbf{v} \in V$, there is a collection of values $c_1, \dots, c_{|V'|}$ for which if $V' = \{\mathbf{v}'_1, \dots, \mathbf{v}'_{|V'|}\}$, then $\mathbf{v} = c_1\mathbf{v}'_1 + \dots + c_{|V'|}\mathbf{v}'_{|V'|}$.

Definition 1.1.E. The *dimension* of V is the smallest integer n such that there exists a spanning set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \subset V$, and such a set is called a *basis*. If there is no such integer n then V is *infinite-dimensional*; otherwise V is *finite-dimensional*. We denote this by $n = \dim V$.

For instance, $\mathbb{R}^3 = \{(a, b, c) \mid a, b, c \in \mathbb{R}\}$ has dimension 3; one basis is $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.

Problem 1.1.1. (2 points) Show that any linearly independent set of size $\dim V < \infty$ in the vector space V over the field F is a basis of V .

Problem 1.1.2. (3 points) Show that $\mathbb{R}[x]$ is a real vector space, and that $\mathbb{C}[x]$ is a complex vector space (where $F[x]$ is the set of finite-degree polynomials with coefficients in F). Is $\mathbb{R}[x]$ a complex vector space? Is $\mathbb{C}[x]$ a real vector space?

1.2 Linear Mappings, Inner Products (11 points)

Definition 1.2.A. A *linear mapping* or *homomorphism* is a function $f : V \rightarrow W$ subject to $f(\mathbf{v} + \mathbf{w}) = f(\mathbf{v}) + f(\mathbf{w})$ and $f(c\mathbf{v}) = cf(\mathbf{v})$ for any $c \in F$ and $\mathbf{v}, \mathbf{w} \in V$, where V and W are vector spaces associated to the same field F .

Definition 1.2.B. In the linear mapping above, V is sometimes called the *domain* of f and W is sometimes called the *range*.

Definition 1.2.C. We define the *image* of f to be $\text{im } f = \{f(\mathbf{v}) \mid \mathbf{v} \in V\} \subseteq W$ and the *kernel* of f to be set $\ker f = \{\mathbf{v} \in V \mid f(\mathbf{v}) = 0\} \subseteq V$.

Consider as an example the linear mapping $f : \mathbb{R} \rightarrow \mathbb{R}^2$ sending $f : x \mapsto (x, 2x)$. $f(cx) = (cx, 2cx) = c(x, 2x) = cf(x)$. This has domain $\mathbb{R}$, range $\mathbb{R}^2$, image $\{(r, 2r) \mid r \in \mathbb{R}\} = \mathbb{R}(1, 2)$, and kernel $\{0\}$.

Definition 1.2.D. A homomorphism $\varphi : V \rightarrow W$ is said to be an *isomorphism* if it has an inverse function $\varphi^{-1} : W \rightarrow V$. V and W are then said to be *isomorphic*.

Definition 1.2.E. A *bilinear mapping* $f : V_1 \times V_2 \rightarrow S$ (where V_1, V_2 are vector spaces over F) satisfies $f(c_1\mathbf{v}_1, c_2\mathbf{v}_2) = c_1c_2f(\mathbf{v}_1, \mathbf{v}_2)$ as well as linearity in both the first and second components.

Definition 1.2.F. An *inner product* is a special type of mapping, which is linear in the first argument (and is “almost linear” in the second); it is denoted $\langle \mathbf{x}, \mathbf{y} \rangle$. $\langle \cdot, \cdot \rangle : V^2 \rightarrow F$ and satisfies:

- $\langle \mathbf{x}, \mathbf{y} \rangle = \overline{\langle \mathbf{y}, \mathbf{x} \rangle}$ (if $F = \mathbb{R}$ then $\bar{r} = r$ for each $r \in \mathbb{R}$)
- $\langle c\mathbf{x}, \mathbf{y} \rangle = c \langle \mathbf{x}, \mathbf{y} \rangle$ for all $c \in F$
- $\langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle = \langle \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle$
- $\langle \mathbf{x}, \mathbf{x} \rangle > 0$ if $\mathbf{x} \neq 0$.

One important inner product is the *standard inner product*, sometimes known as the *dot product*. If $\mathbf{a} = (a_1, a_2, \dots, a_n) \in \mathbb{R}^n$ and $\mathbf{b} = (b_1, b_2, \dots, b_n) \in \mathbb{R}^n$, then $\langle \mathbf{a}, \mathbf{b} \rangle = a_1b_1 + a_2b_2 + \dots + a_nb_n$. You may want to take a minute to verify that this satisfies bilinearity and the inner product conditions.

Problem 1.2.1. (2 points each) Fix $N \in \mathbb{N}$. Let $a(x)$ take the form $a(x) = \sum_{n=0}^N a_n x^n$ and let F_N be the set of degree- N polynomials in $F[x]$. For each of the following functions $f_i : F_N \rightarrow F[x]$, determine with proof whether or not it is a linear map.

- (i) $f_1 : a(x) \mapsto a(x) + a(1)$
- (ii) $f_2 : a(x) \mapsto a(x) + a(1)x^{N+1}$
- (iii) $f_3 : a(x) \mapsto \sum_{n=0}^N (a_n - a_{N-n})x^n$

Problem 1.2.2. (3 points) Describe as precisely as possible all possible inner products on $\mathbb{C}^2$.

Problem 1.2.3. (2 points) If ϕ is an isomorphism, show that ϕ^{-1} is also an isomorphism.

1.3 Matrices (4 points)

Suppose we have the linear map

$$\begin{aligned} f : \mathbf{v}_1 = (1, 0, 0) &\mapsto (2, 0) \\ &: \mathbf{v}_2 = (0, 1, 0) \mapsto (1, 7) \\ &: \mathbf{v}_3 = (0, 0, 1) \mapsto (1, 1). \end{aligned}$$

We can use a shorthand this function: $\begin{bmatrix} 2 & 1 & 1 \\ 0 & 7 & 1 \end{bmatrix}$. The first column represents the image of the first basis vector under the map f ; the same holds for the second and third columns. In general, this format of storing the data of a linear mapping may be used interchangeably with the function itself, since it conveys precisely the same information. This will be important and convenient when it becomes cumbersome to work with the function as a list of operations and instead more useful to use a grid. In general, if $f : V \rightarrow W$ and we fix a basis for each of V and W , then we can represent the map f as a $m \times n$ array of numbers in the following way:

Definition 1.3.A. An $m \times n$ matrix is a grid of values in F having m rows and n columns, where the i th column represents $f(\mathbf{v}_i)$, the image of V 's i th basis vector.

For another example, $\mathbb{R}^2$ has the bases $\{(0, 1), (1, 0)\}$ and $\{(1, 1), (1, -1)\}$. The map $f : (0, 1) \mapsto 4$, $f : (1, 0) \mapsto 2$ has matrix $\begin{bmatrix} 4 & 2 \end{bmatrix}$ with the first basis and $\begin{bmatrix} 6 & -2 \end{bmatrix}$ with the second.

We will denote by A^T the *transpose* of matrix A , having at the j th column and i th row the entry in A 's i th column and j th row. A^T is an $n \times m$ matrix if A is $m \times n$. For instance,

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

We will not treat matrices independently of the context of the functions that they represent.

Matrix addition is rather intuitive; simply add corresponding elements. For example, $\begin{bmatrix} 2 & 0 & 1 & 7 \\ 1 & 7 & 4 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 7 & 3 & 8 \\ 1 & 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 7 & 4 & 15 \\ 2 & 8 & 6 & 8 \end{bmatrix}$.

Multiplication is less obvious, however. When multiplying, we think of the left multiplicand as a column vector of row vectors; for example, $\begin{bmatrix} 2 & 0 & 1 \\ 1 & 7 & 4 \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \end{bmatrix}$ where $\mathbf{v}_1 =$

$\begin{bmatrix} 2 & 0 & 1 \end{bmatrix}$, etc., and the right multiplicand is a row vector of column vectors; for example, $\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 2 & 7 \end{bmatrix} = [\mathbf{w}_1 \quad \mathbf{w}_2]$ where $\mathbf{w}_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$, etc. We simply define the product, then, to be the matrix $\{a_{i,j}\}$ where i ranges along the height of the left multiplicand and j ranges along the width of the right multiplicand, and $a_{i,j} = \mathbf{v}_i \cdot \mathbf{w}_j$. So, $\begin{bmatrix} 2 & 0 & 1 \\ 1 & 7 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 2 & 7 \end{bmatrix} = \begin{bmatrix} 4 & 9 \\ 16 & 29 \end{bmatrix}$. In general, note the rule that a $m \times n$ matrix multiplied by a $n \times p$ matrix is a $m \times p$ matrix.

Definition 1.3.B. The $n \times n$ matrix $I_n = \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & 0 \\ 0 & 0 & 1 \end{bmatrix}$ having 1's along its main diagonal and 0's elsewhere is the *identity* matrix, because $I_n M = M I_n$ for all $n \times n$ matrices M .

Definition 1.3.C. The set of linear mappings between $F^n = F \times F \times \cdots \times F$ and itself is denoted $\mathfrak{gl}_n$.

The elements of $\mathfrak{gl}_n$ can (and often will) be thought of as $n \times n$ matrices. (F is assumed to be $\mathbb{C}$ unless otherwise specified.)

Sometimes for notational convenience we use *block matrices*. For instance, we could write $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ as $\begin{bmatrix} I_2 & 0 \\ 0 & I_2 \end{bmatrix}$, where the 0's actually represent $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$; or we may say

$\begin{bmatrix} I_3 & 0 \\ 0 & 1 \end{bmatrix}$, where one 0 represents $\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$ and the other $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, as to make the size of the matrix's elements make sense.

Problem 1.3.1. (1 point) Does matrix multiplication associate? Why or why not? Answer the same question about commutativity.

The following two problems give that matrix operations and function operations correspond as we expect.

Problem 1.3.2. (1 point) Show that, for linear mappings $T_1, T_2 : \mathbb{R}^m \rightarrow \mathbb{R}^n$ represented by the $n \times m$ matrices $\{a_{i,j}\}$ and $\{b_{i,j}\}$, respectively, the linear mapping $T_1 + T_2 : \mathbb{R}^m \rightarrow \mathbb{R}^n$ can be represented by the $n \times m$ matrix $\{a_{i,j} + b_{i,j}\}$ (where in each instance i ranges along 1 to n and j from 1 to m).

Problem 1.3.3. (2 points) Show that, for matrices $T_1 : \mathbb{R}^n \rightarrow \mathbb{R}^p$ and $T_2 : \mathbb{R}^m \rightarrow \mathbb{R}^n$ and vector $\mathbf{v} \in \mathbb{R}^m$, $T_1(T_2\mathbf{v}) = (T_1 \times T_2)\mathbf{v}$.

Note that all these results also hold for $\mathbb{C}^m$ and $\mathbb{C}^n$, as well as matrices represented as block matrices (though when working with block matrices, it's important to keep careful track of the dimensions of the blocks).

2 Linear Algebra II (22 points)

2.1 Eigen- (7 points)

Suppose we have a linear transformation M of a vector space V and we wish to know M 's fixed points – i.e., the vectors for which $M\mathbf{v} = \mathbf{v}$. Unfortunately, it is not always the case that there exist such vectors, but we can study vectors $\mathbf{v}$ such that $M\mathbf{v} = \lambda\mathbf{v}$ for some $\lambda \in F$.

Definition 2.1.A. If there exists a pair consisting of the field element λ and nonzero vector $\mathbf{v}$ satisfying that relation, then $\mathbf{v}$ is an *eigenvector* and λ is $\mathbf{v}$'s *eigenvalue*.

Definition 2.1.B. A λ -*eigenspace* is the set of elements of V that satisfy $M\mathbf{v} = \lambda\mathbf{v}$. It is often denoted by V_λ (with the linear transformation implied to be already specified).

For instance, $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ has eigenvectors, for any $t \in \mathbb{R}$, $\begin{bmatrix} t \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ t \end{bmatrix}$ with eigenvalues 2 and 1, respectively, and which give $V_2 = \mathbb{R} \times \{0\}$ and $V_1 = \{0\} \times \mathbb{R}$.

Problem 2.1.1. (1 point) Show that each eigenspace V_λ is a subspace of V .

Problem 2.1.2. (1 point) Why does any mapping of a finite-dimensional space have a finite number of distinct eigenvalues?

Problem 2.1.3. (2 points) Suppose $M : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear map with matrix $\begin{bmatrix} 2 & 0 \\ 1 & 7 \end{bmatrix}$. Find each eigenvector and the corresponding eigenvalue of M . (Hint: which linear map sends the vector $\mathbf{v}$ to $c\mathbf{v}$ ($c \in \mathbb{R}$)?)

Problem 2.1.4. (3 points) For each function in Problem 1.2.1, if it is a linear map, determine its eigenvalues and classify as best you can the associated eigenvectors.

2.2 Trace (12 points)

Definition 2.2.A. The *trace* of A , an $n \times n$ square matrix having value $a_{i,j}$ at column i and row j , is equal to $\sum_{k=1}^n a_{k,k}$, that is, the sum of its diagonal entries.

For instance, $\text{tr} \begin{bmatrix} 2 & 0 \\ 1 & 7 \end{bmatrix} = 2 + 7 = 9$.

Lemma 2.2.1. $\text{tr}(XY) = \text{tr}(YX)$ for X, Y $n \times n$ matrices.

This result is very useful; for practice, you may try to prove it.

Definition 2.2.B. $\mathfrak{sl}_n$ is the set of $n \times n$ complex matrices with trace 0.

While we're on the topic of defining subspaces of $\mathfrak{gl}_n$, let's introduce two more classes of matrices. Denote by $R_{2n} = \begin{bmatrix} 0 & I_n \\ I_n & 0 \end{bmatrix}$, by $R_{2n+1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & I_n \\ 0 & I_n & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & R_{2n} \end{bmatrix}$, and by $\tilde{R}_{2n} = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}$.

Definition 2.2.C. $\mathfrak{so}_n$ is the set of $n \times n$ complex matrices M that satisfy $M^T R_n + R_n M = 0$.

Definition 2.2.D. $\mathfrak{sp}_{2n}$ is the set of $2n \times 2n$ complex matrices M that satisfy $M^T \tilde{R}_{2n} + \tilde{R}_{2n} M = 0$. (We do not define $\mathfrak{sp}_{2n+1}$.)

While these definitions may come across as unmotivated, we will eventually discover some of their remarkable properties.

Problem 2.2.1. (2 points) What is $\dim \mathfrak{sl}_n$? Find a basis.

Problem 2.2.2. (10 points) Classify as best you can the elements of $\mathfrak{so}_n$ and $\mathfrak{sp}_n$; find $\dim \mathfrak{so}_n$ and $\dim \mathfrak{sp}_n$.

2.3 Semisimplicity & nilpotency (3 points)

Definition 2.3.A. A *diagonal matrix* is one where each of its entries not on the main diagonal are zero.

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ are both examples of diagonal matrices.

Suppose, for the rest of this section, that all matrices considered lie in $\mathfrak{gl}_n$ for a fixed n .

Definition 2.3.B. The matrix A is *semisimple* if there is diagonal matrix D and another matrix P having a multiplicative inverse such that $PA = DP$.

Clearly, all diagonal matrices are semisimple, by choosing $P = I$. Additionally, it is important to note that the invertibility of P is essential, since otherwise any matrix would be semisimple, by choosing $P = 0$.

Definition 2.3.C. The matrix N is *nilpotent* if there is a positive integer k for which $N^k = 0$.

The matrix 0 is trivially nilpotent. A less obvious example is $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ (it becomes 0 after squaring).

It turns out that semisimple and nilpotent matrices have a very deep relationship.

Problem 2.3.1. (3 points) Show that if X and Y are matrices that commute and X is nilpotent then $\text{tr}(XY) = 0$.

We are now ready to begin investigating Lie algebras. Despite their simple definition, we will soon see a host of powerful results that aid in their classification.

3 Lie Algebras I (77 points)

3.1 What is a Lie algebra? (26 points)

Definition 3.1.A. A *Lie algebra* is a vector space L with, besides addition, the bilinear operation $[\cdot, \cdot] : L \times L \rightarrow L$. This *bracket* (sometimes the *Lie bracket*) is subject to the following properties:

- $[x, x] = 0$ for $x \in L$
- $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$ for any $x, y, z \in L$ (the *Jacobi identity*)

Linear maps from $F^n \rightarrow F^n$ are actually a very good way to think about many Lie algebras. (You'll show in Problem 3.1.1 that these functions do indeed form a Lie algebra.)

Lemma 3.1.1 (*anticommutativity of the bracket*). $[x, y] = -[y, x]$ for x, y elements of a Lie algebra L .

Proof. Let $z = x + y$. $0 = [z, z] = [x + y, x + y] = [x, x] + [x, y] + [y, x] + [y, y] = [x, y] + [y, x]$. $\square$

Definition 3.1.B. A Lie algebra for which the bracket is degenerate, i.e. $[x, y] = 0$ for all $x, y \in L$, is termed an *abelian* Lie algebra.

Note that the *dimension* of a Lie algebra is its dimension as a vector space under addition. In general, we do not have a meaningful way to treat the bracket operation as an operation giving rise to a vector space.

We also have a natural extension to the concept of vector space homomorphism and isomorphism:

Definition 3.1.C. A *homomorphism* of Lie algebras is a function $\varphi : L_1 \rightarrow L_2$ where L_1 and L_2 are Lie algebras over the same field, and for which $\varphi([x, y]) = [\varphi(x), \varphi(y)]$ for any $x, y \in L_1$.

Definition 3.1.D. An *isomorphism* of Lie algebras is a bijective homomorphism of Lie algebras.

The following problems will serve to hone your intuition.

Problem 3.1.1. (1 point) Let $[f, g] = f \circ g - g \circ f$ for $f, g \in \mathfrak{gl}_n$. Show that the Jacobi identity holds for $f, g, h \in \mathfrak{gl}_n$, and therefore that $\mathfrak{gl}_n$ is a Lie algebra.

Problem 3.1.2. (1 point) Explain why two Lie algebras of different dimension cannot be isomorphic.

Problem 3.1.3. (4 points) Show that if L_1 and L_2 are abelian Lie algebras, they are isomorphic if and only if they are of the same dimension.

Problem 3.1.4. (4 points) Assume that for all $x, y, z \in L$, $[x, [y, z]] = [[x, y], z]$. What is the most general statement you can make about elements of L ?

Problem 3.1.5. Fix n . Suppose there is some subalgebra $S \subseteq \mathfrak{gl}_n$ for which $\phi : \mathfrak{sl}_2 \rightarrow S$ is a bijective Lie algebra homomorphism (where $\mathfrak{gl}_n$ has the bracket as in Problem 3.1.1). If $v_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, let $V_i = \phi(v_i)$. Let λ_3 and λ'_3 be the greatest and least eigenvalue, respectively, of V_3 and let $v, v' \in \mathbb{C}^n$ be vectors such that $V_3 v = \lambda_3 v$ and $V_3 v' = \lambda'_3 v'$.

(i) (5 points) Find, with proof, $V_1 v$ and $V_2 v'$.

(ii) (7 points) Find, with proof, λ_3 and λ'_3 .

Problem 3.1.6. (4 points) Up to isomorphism, how many Lie algebras have dimension 1? dimension 2? List them all.

3.2 Ideals and Subalgebras (16 points)

Definition 3.2.A. A *subalgebra* S of the Lie algebra L is a subspace for which if x and y are any elements of S then $[x, y] \in S$.

Definition 3.2.B. An *ideal* I is a subalgebra for which if x is any element of I and y is any element of L then $[x, y] \in I$.

One example is the x -axis in $\mathbb{R}^2$ (i.e. $\{(x, 0) \mid x \in \mathbb{R}\}$ under the bracket $[P, Q] = 0$. The axis is a vector space, and thus both a subalgebra and an ideal.

Definition 3.2.C. A Lie algebra L is *simple* if its only ideals are $\{0\}$ and itself.

One example of a simple Lie algebra is $\mathfrak{sl}_2$. (To prove this, you can show that no subspace of dimension 1 or 2 may be an ideal, by forcing contradictions by bracketing with the right choices of v_1, v_2, v_3 .)

Problem 3.2.1.

- (i) (1 point) If L is an abelian Lie algebra, show that *any* subspace of L is a subalgebra and an ideal.
- (ii) (4 points) Find a nonabelian Lie algebra on a space of $n \times n$ matrices for some $n > 2$ that has a subalgebra that is not an ideal. Specify that subalgebra and briefly justify why it is not an ideal.

Problem 3.2.2. (5 points) Take $I = \{x \in L \mid x \text{ nilpotent}\}$ where L is a subspace of $\mathfrak{gl}_n$. Is I a subalgebra? an ideal? Fully justify your responses.

Problem 3.2.3. (3 points) Let $[L_1, L_2] = \{[v_1, v_2] \mid v_i \in L_i\}$ for Lie algebras L_1 and L_2 . Is $[I, J]$ an ideal for ideals I and J of the same Lie algebra?

Problem 3.2.4. (3 points) Show that $\ker \phi$ is an ideal and $\text{im } \phi$ is a subalgebra of the domain and image space, respectively, for ϕ a Lie algebra homomorphism.

3.3 Ado's Theorem (9 points)

Theorem 3.3.1 (*Ado's theorem*). For any finite-dimensional Lie algebra L , there exists a positive integer n such that there is a subalgebra of $\mathfrak{gl}_n$ that is isomorphic to L .

Its proof is well beyond the scope of this Power Round, so we will assume it to hold true. **We will also assume that any Lie algebra henceforth is a matrix subalgebra**, i.e. a subalgebra of $\mathfrak{gl}_n$ for some n . For any matrix space, we define the bracket to take $[X, Y] = XY - YX$; since the matrices are linear mappings, note that this is the definition we have used earlier, such as in Problem 3.1.1.

In general it is not strictly important *what* the isomorphism is between a non-matrix Lie algebra and the corresponding matrix subalgebra, but to further your comfort with isomorphisms and the Lie bracket, this section's exercises will be concerned with that task.

Problem 3.3.1. Consider the additive vector space V_d of degree- d polynomials with real coefficients.¹

¹Note that $x^3 + x + 1$ is a polynomial of degree 3, 4, 5, etc. but **not** 2, 1 or 0.

- (i) (1 point) Find the dimension of V_3 , and find a basis.
- (ii) (5 points) Say V_2 is given a bracket operation, for $p(x) = \alpha_0 + \alpha_1x + \alpha_2x^2$ and $q(x) = \beta_0 + \beta_1x + \beta_2x^2$ elements of V_2 ,

$$[p, q] = (\alpha_1\beta_2 - \alpha_2\beta_1) + (\alpha_2\beta_0 - \alpha_0\beta_2)x + (\alpha_0\beta_1 - \alpha_1\beta_0)x^2.$$

Find a subspace S of $\mathfrak{gl}_n$ for some n such that there is an isomorphism between V_2 and S ; explicitly describe such an isomorphism that preserves the bracket.

Problem 3.3.2. (3 points) Suppose $\mathbb{R}^2$ is given a Lie bracket that makes it abelian. Find a subspace S of $\mathfrak{gl}_n$ for some n such that there is an isomorphism between $\mathbb{R}^2$ and S ; explicitly describe the isomorphism.

3.4 The adjoint representation (26 points)

Definition 3.4.A. The *adjoint representation* of $x \in L$ is a function taking L to itself in the following manner: $\text{ad } x : y \mapsto [x, y]$.

In practice, we write the action of x 's adjoint representation on y as $(\text{ad } x)(y) = [x, y]$.

Problem 3.4.1. (1 point) Verify that $\text{ad } x$ is a linear map.

Problem 3.4.2.

- (i) (2 points) Explain how $\text{ad } x$ can be thought of as an element of $\mathfrak{gl}_3$ if $x \in \mathfrak{sl}_2$.
- (ii) (5 points) Find an explicit, nontrivial² isomorphism from the Lie algebra $\mathfrak{sl}_2$ to a subalgebra of $\mathfrak{gl}_3$.

Problem 3.4.3. (5 points) Show that the set of 3×3 complex upper-triangular matrices is a Lie algebra, find a basis, and write $\text{ad } e_i$ explicitly for each basis vector e_i . Find $\text{ad } e_i$'s eigenvalues for each basis vector e_i .

Hint for 3.4.2(ii) and 3.4.3: Remember that $\text{ad } x$ is a linear map from L to itself, or equivalently, from S to itself, where $S \subseteq \mathfrak{gl}_n$ is isomorphic to L . Also, make sure you specify the order of the elements of your basis when writing your matrices.

Problem 3.4.4. (5 points) Show that $[x, y] = 0$ implies $(\text{ad } x) \circ (\text{ad } y) = (\text{ad } y) \circ (\text{ad } x)$.

Problem 3.4.5. (6 points) If x is nilpotent, is $\text{ad } x$ nilpotent? If $\text{ad } x$ is nilpotent, is x nilpotent? Justify your response.

Problem 3.4.6. (2 points) Show that $\text{ad } x$ satisfies the relation $(\text{ad } x)([y, z]) = [(\text{ad } x)(y), z] + [y, (\text{ad } x)(z)]$ for any $x, y, z \in L$. (Such functions are called *derivations*.)

²e.x., $M \mapsto \begin{bmatrix} M & 0 \\ 0 & 0 \end{bmatrix}$ is such a trivial mapping. The problem isn't very much fun if read to allow trivial solutions, is it?

4 Lie Algebras II (55 points)

4.1 The Killing form and semisimple Lie algebras (24 points)

Definition 4.1.A. The *Killing form* is an operation defined as

$$\kappa(x, y) = \text{tr}(\text{ad } x \cdot \text{ad } y).$$

(Don't be intimidated by the name!)

Definition 4.1.B. We call κ *non-degenerate* and we say that L is a *semisimple* Lie algebra if the only element $x \in L$ for which $\kappa(x, y) = \text{tr}(\text{ad } x \cdot \text{ad } y) = 0$ for all y is $x = 0$.

As it turns out, all the Lie algebras that we have considered thus far have been semisimple. We now restrict our attention to such algebras, and with this definition to guide us, we are about to see a number of remarkable results.

Problem 4.1.1. (1 point) Verify that the Killing form is symmetric and bilinear.

Problem 4.1.2. (5 points) Show that $\kappa([x, y], z) = \kappa(x, [y, z])$.

Problem 4.1.3. (10 points) Show that $x \in L$ is semisimple if and only if $\text{ad } x$ is semisimple.

Problem 4.1.4.

- (i) (3 points) Let $I' = \{x \in L \mid \kappa(x, y) = 0 \forall y \in I\}$ for $I \subseteq L$ an ideal. Show that I' is also an ideal.
- (ii) (5 points) If L is semisimple, show that there are n simple subalgebras L_i ($1 \leq n \leq \dim L$) such that

$$L = \bigoplus_{i=1}^n L_i.$$

4.2 Root space decomposition (31 points)

Definition 4.2.A. The *center* of a subalgebra $S \subseteq L$ is $\{x \in L \mid [x, y] = 0 \forall y \in S\}$.

Theorem 4.2.1. Any Lie algebra L has a nontrivial subalgebra H of semisimple elements that is equal to its center in L ; H is known as the *maximal toral subalgebra*. There exists a basis of L such that each basis element is an eigenvector to each element of H .

In particular $[h, v_i] = \lambda_i(h)v_i$ for some function $\lambda_i : H \rightarrow \mathbb{C}$. For example, for the sub-basis spanning H consisting of $\{v_1, \dots, v_{\dim H}\}$ (renumbered as necessary), $\lambda_i(h) = 0$ identically for $1 \leq i \leq \dim H$, since H is equal to the center, where the bracket is abelian (but this would not necessarily hold for an element not in H).

Definition 4.2.B. λ_i is called a *root* of L , and H^* is the vector space with spanned by $\{\lambda_i \mid 1 \leq i \leq \dim L\}$.

Definition 4.2.C. The *root space* L_λ corresponding to the root λ consists of the elements of L that are λ -eigenvectors of H – i.e.,

$$L_\lambda = \{v \in L \mid [h, v] = \lambda(h)v \forall h \in H\}.$$

This yields the *root space decomposition* of L .

From each root λ we then are able to derive a convenient isomorphism: there are three elements $e_\lambda \in L_\lambda$, $f_\lambda \in L_{-\lambda}$ and $h_\lambda \in H$ for which $\mathfrak{sl}_2$ is isomorphic to $\text{span}\{e_\lambda, f_\lambda, h_\lambda\}$ via

$$\begin{aligned}\varphi : v_1 &\mapsto e_\lambda \\ v_2 &\mapsto f_\lambda \\ v_3 &\mapsto h_\lambda\end{aligned}$$

and the additional condition that $\lambda(h_\lambda) = 2$.

We are now also ready to talk about H^* as a vector space having an inner product.

Definition 4.2.D. $t_\lambda \in H$ corresponds to $\lambda \in H^*$ such that $\kappa(t_\lambda, h) = \lambda(h)$ for each $h \in H$.

Then, we get a convenient inner product for our space space of functions! We let the inner product on H^* be $\langle \lambda_1, \lambda_2 \rangle = \kappa(t_{\lambda_1}, t_{\lambda_2})$.

Problem 4.2.1. (5 points) Show that each λ_i is linear.

Problem 4.2.2. (3 points) Show that L is isomorphic to the vector space

$$\bigoplus_{\lambda: H \rightarrow \mathbb{C}} L_\lambda.$$

Problem 4.2.3. (6 points) Suppose α and β are roots of L having maximal toral subalgebra H . Under what condition on α and β is it true that $\kappa(v_1, v_2) \neq 0$ for some $v_1 \in L_\alpha, v_2 \in L_\beta$?

Hint: you may not be able to prove the result stated in this exact manner.

Hint 2: try multiplying by a nice factor.

Problem 4.2.4. (6 points) Show that $\langle \cdot, \cdot \rangle : H^* \times H^* \rightarrow \mathbb{C}$ as defined above is an inner product.

Problem 4.2.5. (6 points) Show that $[x, y] = \kappa(x, y)t_\lambda$ for $x \in L_\lambda, y \in L_{-\lambda}$.

Problem 4.2.6. (5 points) Show that for roots $\lambda \in H^*$, $\kappa(t_\lambda, t_\lambda)\kappa(h_\lambda, h_\lambda) = 4$ and show that $h_\lambda = \frac{2t_\lambda}{\kappa(t_\lambda, t_\lambda)}$.

5 Root systems (125 points)

5.1 What is a root system? (37 points)

We will briefly step back to consider any vector space, which might not have a Lie bracket.

Definition 5.1.A. A subset (**note:** not a subspace) S of a real vector space V with the inner product $\langle \cdot, \cdot \rangle$ is a *root system* of V if $\text{span } S = V$ and, for any $v, w \in S$,

1. the only other scalar multiple of v in S is $-v$

2. $w - 2\frac{\langle v, w \rangle}{\langle v, v \rangle}v \in S^3$
3. $2\frac{\langle v, w \rangle}{\langle v, v \rangle} \in \mathbb{Z}$

Elements of the root system are known as *roots*. (This is related to the definition in Section 4.3.)

Note that a root system is not guaranteed to exist.

Definition 5.1.B. A *reducible* root system may be decomposed into nonempty subsets $R_1 \cup R_2$ for which $\langle v_1, v_2 \rangle = 0$ for $v_1 \in R_1, v_2 \in R_2$, where R_1 and R_2 are also root systems. An *irreducible* root system is one that is not reducible.

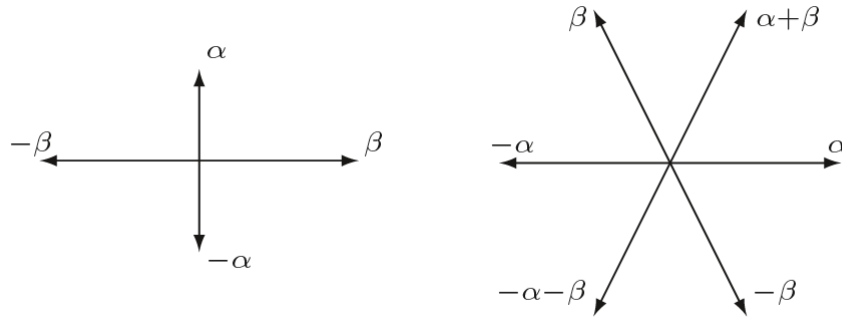


Figure 1: Two root systems of $\mathbb{R}^2$, (a) $A_1 \times A_1$ and (b) A_2 .

Figure 1(a) depicts the root system of $\mathbb{R}^2$

$$R_{1a} = \{[2 \ 0], [-2 \ 0], [0 \ 1], [0 \ -1]\}.$$

Note that $R_{1a} = \{[2 \ 0], [-2 \ 0]\} \cup \{[0 \ 1], [0 \ -1]\}$ and that

$$\text{span } R_{1a} = \mathbb{R}^2 = \text{span } \{[2 \ 0], [-2 \ 0]\} \oplus \text{span } \{[0 \ 1], [0 \ -1]\}.$$

We see that $A_1 \times A_1$ is a reducible root system. Meanwhile, Figure 1(b) depicts the irreducible root system of $\mathbb{R}^2$

$$R_{1b} = \left\{ [1 \ 0], [-1 \ 0], \left[\frac{1}{2} \ \frac{\sqrt{3}}{2} \right], \left[-\frac{1}{2} \ \frac{\sqrt{3}}{2} \right], \left[-\frac{1}{2} \ -\frac{\sqrt{3}}{2} \right], \left[\frac{1}{2} \ -\frac{\sqrt{3}}{2} \right] \right\};$$

note that only trivially can R_{1b} be broken up into smaller root spaces (i.e. via $\{0\}$ and R_{1b} itself).

Definition 5.1.C. R has *base* $B \subset R$ if B is a basis of V and, for each $w \in R$, there exists $c : B \rightarrow \mathbb{N} \cup \{0\}$ such that $w = \pm \sum_{v \in B} c(v)v$.

³This represents reflecting w across the plane in V perpendicular to v and passing through 0.

Going back to our examples, $A_1 \times A_1$ can have the base $\left\{ \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$ and A_2 can have the base $\left\{ \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \right\}$.

Problem 5.1.1. (1 point) Find all root systems in $\mathbb{R}$.

Problem 5.1.2. (3 points) In condition 3, note that v and w are interchangeable. Use this and the identity that $\langle v, w \rangle = \|v\| \cdot \|w\| \cos \phi$ (where ϕ is the angle between the vectors) to show that $4 \cos^2 \phi \in \{0, 1, 2, 3\}$ whenever $v \neq \pm w$.

Problem 5.1.3. (8 points) Think about the construction of the root systems in the text, as well as Problem 5.1.2, use these to understand why the subsets of $\mathbb{R}^2$ shown in Figure 1 are root systems. Draw (on separate coordinate axes) all possible root systems R .⁴

Problem 5.1.4. (10 points) Show that any root system can be written as the union of irreducible root systems. (Use the fact that a root system R is irreducible if there is no subset R' such that for $\alpha \in R'$ and $\beta \in R \setminus R'$, $\langle \alpha, \beta \rangle = 0$.)

Problem 5.1.5. (15 points) Show that any root system has a base.

5.2 Dynkin diagrams (48 points)

Let B be a base of a root system and let $B' = \left\{ \frac{v}{\sqrt{\langle v, v \rangle}}, v \in B \right\}$ (we *normalize* the base).

We now assume that B' satisfies $\langle v_i, v_j \rangle \leq 0$ when $i \neq j$, and $4\langle v_i, v_j \rangle^2 \in \{0, 1, 2, 3\}$.

We will represent the normalized base as a graph with *multi-edges*, i.e. a number of edges, possibly greater than one (and in that case directed), and possibly zero, connecting two given vertices. We will think of these as “total edges,” so that there cannot simultaneously be edges going in both directions. We define this as

$$e(v, w) = 4 \frac{\langle v, w \rangle^2}{\|v\|^2 \|w\|^2}.$$

(Note that this is equal to $\left(2 \frac{\langle v, w \rangle}{\sqrt{\langle v, v \rangle}}\right) \cdot \left(2 \frac{\langle w, v \rangle}{\sqrt{\langle w, w \rangle}}\right)$.)

Since we have an explicit characterization of the nature of edges connecting any two simple roots, we simply draw a node for each of those roots and the appropriate edge to each other simple root. Frequently an edge will be zero, so we simply will not draw an edge at all; otherwise, we draw a straight, undirected edge between v and w if $e(v, w) = 1$ and a directed n -fold edge if $e(v, w) = n$. The question, then, is how to determine the direction of the edge, since e is symmetric in its inputs v and w ($e(v, w) = e(w, v)$). Our resolution is simply to point from the longer root to the shorter one, where the length of v is defined as $\ell = \sqrt{\langle v, v \rangle}$.⁵

As an example, let's find the diagrams for $A_1 \times A_1$ and A_2 . We will use $B'_1 = \{w_1 = (1, 0), w_2 = (0, 1)\}$ and $B'_2 = \{v_1 = (1, 0), v_2 = (-\frac{1}{2}, \frac{\sqrt{3}}{2})\}$. We are using the dot product for $\langle \cdot, \cdot \rangle$. $e(w_1, w_2) = 0$ and $e(v_1, v_2) = 1$, so we have the following Dynkin diagrams:

⁴Up to rotation and dilation; that is, if R_1 and R_2 are two root systems that you submit, there should not exist a nonidentity matrix M for which $R_1 = MR_2$.

⁵One can check that for the standard inner product on $\mathbb{R}^n$ this is equivalent to the regular conception of length using the distance formula.



Figure 2: Dynkin diagram for $A_1 \times A_1$ on the left (the two disconnected points) and A_2 on the right.

Before continuing onto the Problems, let's look at a sample proof to gain a sense for Dynkin diagram arguments. Consider a line which may be part of a larger Dynkin diagram.



Figure 3: A line (sub)diagram.

Now consider “combining” these into a “super-vertex” by “folding along” – i.e., let $v = v_1$; then let $v = v + v_2$; then let $v = v + v_3$; etc., until we get a single vector $v = \sum_{i=1}^k i v_i$. We will show that $\langle v, v \rangle = \frac{k(k+1)}{2}$.

Proof. We know that $\langle v_i, v_j \rangle \neq 0$ iff $|i - j| \leq 1$. $\langle v_i, v_i \rangle = 1$ since they are unit vectors, and $2\langle v_i, v_{i+1} \rangle = -1$ since the edge length is 1, and the inner product here cannot be positive.⁶ Thus $\langle v, v \rangle = \sum_{i=1}^k i^2 + 2 \sum_{i=1}^{k-1} \langle v_i, v_{i+1} \rangle i(i+1) = \frac{k(k+1)}{2}$ by computation. $\square$

This should serve as a viable model for the coming problems that ask about the impossibility of certain Dynkin diagrams – look to use the fact that the number of edges is necessarily a nonnegative integer.

Problem 5.2.1. (10 points) Show that a root system is irreducible if and only if it has a connected Dynkin diagram.

Problem 5.2.2. (7 points) Show that no Dynkin diagram may have a cycle.

Problem 5.2.3. (8 points) Show that no node in a Dynkin diagram may have more than three total edges connected to it.

We'll now restrict our consideration to only connected Dynkin diagrams.

Problem 5.2.4. (4 points) The *Shrinking Lemma* states that if a Dynkin diagram on the base B has n vertices $v_1, v_2, \dots, v_n$ that are in a *line* and $v = \sum_{i=1}^n v_i$, then the base $B \setminus \{v_1, v_2, \dots, v_n\} \cup \{v\}$ corresponds to the root system with the intermediate vertices “combined” into a single vertex (as before). (See Figure 4.) Use the Shrinking Lemma⁷ to show

⁶Think about why; this will be useful in the coming problems. Hint: what would be wrong with these vectors having an acute angle? What would that imply?

⁷The Shrinking Lemma is not exceptionally difficult to prove, but it is something of a digression. You may want to try proving it to test your command of the material, but henceforth you may assume it is true.

that no Dynkin diagram may have more than one branch, more than one double-edge, or both a branch and a double edge.



Figure 4: An impossible Dynkin diagram. The Shrinking Lemma gives that if there is a root system corresponding a variant of this diagram (with n nodes amongst the $\dots$) then there is one corresponding to the one with $n - 1, n - 2, \dots, 1, 0$ vertices.

Problem 5.2.5. (4 points) Prove that if a Dynkin Diagram has a double edge with both ends connected to other nodes then it must be F_4 .

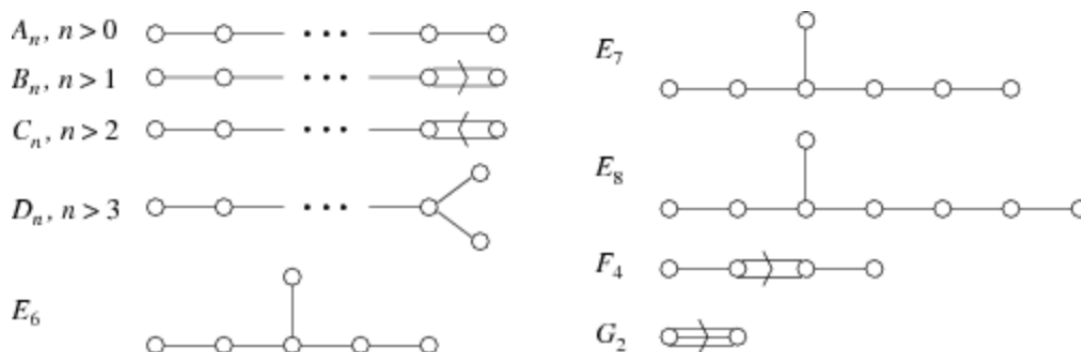


Figure 5: F_4 , the Dynkin diagram discussed in Problem 5.3.6.

Problem 5.2.6. (15 points) Imagine a Dynkin diagram with a “branching point” – there is some vector with three lines coming off of it. Show that one of those lines must have length one (i.e. only one edge), and show that of the other two, either

- if one has length 1 then the third may have any length
- if one has length 2 then the other’s length may not be more than 4.

Theorem 5.2.1 (*Classification of Dynkin diagrams*). The following are the only possible irreducible Dynkin diagrams:



Proof. Problems 5.2.1 and 5.2.3-7. □

This is an amazing result! We gave only a handful of restrictions on our definition of these diagrams – and yet we get such a restricted and asymmetrical list of options. Let’s now look into how this relates to the Lie algebra content that we spent so much time building up.

5.3 Dynkin diagrams of Lie algebras (40 points)

We are now prepared to look at root systems and bases of our simple Lie algebras! In doing so, with a little help from Jean-Pierre Serre, we will be able to see, pictorially, precisely every isomorphism between the classical Lie algebras.⁸

Definition 5.3.A. The Dynkin diagram of a Lie algebra is the Dynkin diagram corresponding to the vector space of its roots.

Definition 5.3.B. $e_{i,j}$ denotes the $N \times N$ matrix having the entry 1 at row i and column j , and zeros elsewhere (where the value of N will be clear from context).

For a worked example, take $L = \mathfrak{so}_{2n}$. As L 's elements satisfy $x^T R_{2n} + R_{2n} x = 0$, we know that x can be written in the form $\begin{bmatrix} M & P \\ Q & -M^T \end{bmatrix}$ for any $n \times n$ matrix M and $n \times n$ matrices P and Q both equal to their negative transpose.

H is the set of diagonal matrices in L , so $H = \left\{ \sum_{i=1}^n c_i (e_{i,i} - e_{i+n,i+n}) \mid c_i \in \mathbb{C} \right\}$. We can give L the additional basis elements $m_{i,j} = e_{i,j} - e_{j+n,i+n}$ ($i \neq j$), $p_{i,j} = e_{i,j+n} - e_{j,i+n}$ ($i < j$), and $p_{i,j}^T$ to go along with those of H . (It is straightforward to check that this is indeed a basis of L – try writing down the matrices $m_{i,j}$ and $p_{i,j}$ to verify this.) We find, for an arbitrary element $h \in H$ of the form specified in H 's specification,

$$\begin{aligned} [h, m_{i,j}] &= (c_i - c_j) m_{i,j} \\ [h, p_{i,j}] &= (c_i + c_j) p_{i,j} \\ [h, p_{i,j}^T] &= -(c_i + c_j) p_{i,j}^T \end{aligned}$$

so our roots are particularly convenient:

root	eigenspace
$\lambda_i - \lambda_j$	$\text{span} \{m_{i,j}, m_{j,i}\}$
$\lambda_i + \lambda_j$	$\text{span} \{p_{i,j}, p_{j,i}\}$

For H^* , a vector space having the basis $\{\lambda_i\}$, we see that $\{\lambda_1 - \lambda_2, \lambda_2 - \lambda_3, \dots, \lambda_{n-1} - \lambda_n, \lambda_{n-1} + \lambda_n\}$ is a base; this base will be what we use to build our Dynkin diagram. Computation gives that

$$\begin{aligned} \langle \lambda_i - \lambda_{i+1}, \lambda_j - \lambda_{j+1} \rangle &= \begin{cases} 2 & i = j \\ -1 & |i - j| = 1 \\ 0 & \text{otherwise} \end{cases} \\ \langle \lambda_i - \lambda_{i+1}, \lambda_{n-1} + \lambda_n \rangle &= \begin{cases} -1 & i = n - 2 \\ 0 & \text{otherwise} \end{cases} \\ \langle \lambda_{n-1} + \lambda_n, \lambda_i - \lambda_{i+1} \rangle &= \begin{cases} -1 & i = n - 2 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Since $e(v, w) = \langle v, w \rangle \cdot \langle w, v \rangle$, our Dynkin diagram is shown in Figure 6.

⁸The classical Lie algebras are the four spaces defined in Section 2.2: $\mathfrak{sl}_n, \mathfrak{so}_n, \mathfrak{sp}_{2n}$.

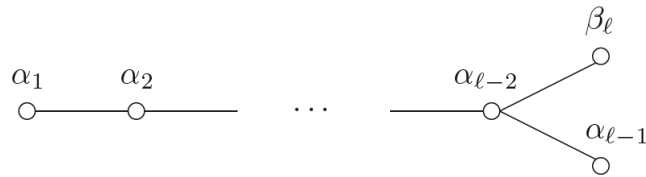


Figure 6: In this diagram, we use the shorthands $\alpha_i = \lambda_i - \lambda_{i+1}$ and $\beta_n = \lambda_{n-1} + \lambda_n$.

Theorem 5.3.1 (*Serre's theorem*). Two Lie algebras with the same Dynkin diagram are isomorphic.

Serre's theorem is the final piece of the puzzle; now, you will put to the test what we have learned to find the bases of the other classical Lie algebras.

Problem 5.3.1. (10 points each) For $n \geq 1$ in all cases...

- (i) draw the Dynkin diagram for $\mathfrak{so}_{2n+1}$
- (ii) draw the Dynkin diagram for $\mathfrak{sl}_n$
- (iii) draw the Dynkin diagram for $\mathfrak{sp}_{2n}$

and show your reasoning. In particular, many details of the computations were omitted above, but you will be expected to (briefly) justify the numbers you find.

What you should get from this is that we have discussed exactly four infinite classes of classical Lie algebras, and there are exactly four infinite classes of Dynkin diagrams. This, combined with Serre's theorem, takes us to our last result.

Problem 5.3.2. (10 points) State all isomorphisms between the classical Lie algebras, and the Dynkin diagrams that you used to reach those conclusions. (Do not provide the isomorphisms explicitly – just briefly state your reasoning.)

That's it! We've managed to find all the equivalencies between the classical Lie algebras, admittedly in a rather roundabout manner – but I hope you've enjoyed the scenery along the way!

Team Number: _____

PUMaC 2017 Power Round Cover Sheet

Remember that this sheet comes first in your stapled solutions. You should submit solutions for the problems in increasing order. Write on one side of the page only. The start of a solution to a problem should start on a new page. Please mark which questions for which you submitted a solution to help us keep track of your solutions.

Problem Number	Points	Attempted?
1.1.1	2	
1.1.2	3	
1.2.1	6	
1.2.2	3	
1.2.3	2	
1.3.1	1	
1.3.2	1	
1.3.3	2	
2.1.1	1	
2.1.2	1	
2.1.3	2	
2.1.4	3	
2.2.1	2	
2.2.2	10	
2.3.1	3	
3.1.1	1	
3.1.2	1	
3.1.3	4	
3.1.4	4	
3.1.5	12	
3.1.6	4	
3.2.1	5	
3.2.2	5	
3.2.3	3	
3.2.4	3	
3.3.1	6	
3.3.2	3	
3.4.1	1	
3.4.2	7	

Problem Number	Points	Attempted?
3.4.3	5	
3.4.4	5	
3.4.5	6	
3.4.6	2	
4.1.1	1	
4.1.2	5	
4.1.3	10	
4.1.4	8	
4.2.1	5	
4.2.2	3	
4.2.3	6	
4.2.4	6	
4.2.5	6	
4.2.6	5	
5.1.1	1	
5.1.2	3	
5.1.3	8	
5.1.4	10	
5.1.5	15	
5.2.1	10	
5.2.2	7	
5.2.3	8	
5.2.4	4	
5.2.5	4	
5.2.6	15	
5.3.1	30	
5.3.2	10	
Total	299	